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Editors
PLOS Computational Biology

Dear Editors,

I am submitting “A Pre-Registered Direction Instability Atlas Across Three Perturbation Modalities” for consideration as a Research Article in *PLOS Computational Biology*.

Whether a perturbation produces consistent effects across cell types determines if laboratory findings transfer to patients, yet no existing metric separates inconsistent effects from merely weak ones. A companion paper under review at this journal (PCOMPBIOL-S-26-02155) introduced direction instability (DI) and validated it on LINCS L1000 drug data. This paper extends DI across three measurement technologies—L1000 transcriptomics, Tahoe-100M single-cell expression, and JUMP Cell Painting morphology—covering ~55,000 drugs and genetic perturbations.

Eleven pre-registered hypothesis tests reveal that DI is structured by pharmacological mechanism: drugs targeting core cellular machinery act consistently across cell types, while receptor-signaling drugs produce context-dependent effects. Essential genes produce stereotyped disruption regardless of cell type. A highly selective drug can still have high DI if its target pathway is wired differently across contexts—selectivity alone does not predict transferability. Eight of eleven tests fail their pre-registered criteria, delimiting what the metric captures and preventing overclaiming.

All data, code, pre-registration documents, and results are deposited on Zenodo (DOI: 10.5281/zenodo.21223667). I am the sole author and declare no competing interests. The manuscript is not under consideration at another journal.

Thank you,

Elliot Tower